

Log Scrubber - User Guide

`log_scrubber.py` is a standalone command-line utility and importable Python library for scrubbing CUI, PII, PHI, credentials, and other sensitive data from Splunk field-value exports and log samples.

This guide is the CLI-focused quick-start. For the comprehensive project documentation covering both the CLI and the Paydirt browser tool, see the [README.md](#) at the repository root.

Compliance coverage: CMMC (CUI marking detection per NIST SP 800-171), HIPAA (PHI identifiers including SSN, Luhn-validated payment cards, NPI, phone), GDPR (personal data including IPs and device identifiers).

No dependencies beyond Python 3.9+ standard library.

Quick Start

```
log_scrubber/  
├─ log_scrubber.py  
├─ log_scrubbing_config.csv    <- place your config here  
└─ (your input files)
```

```
# Auto-detect mode from file contents (easiest)  
python log_scrubber.py my_fields.csv  
  
# Batch multiple files - mode auto-detected per file  
python log_scrubber.py ~/Downloads/*.csv  
  
# Explicit mode (still supported)  
python log_scrubber.py fieldsummary my_fields.csv  
python log_scrubber.py samples my_events.csv  
  
# Include raw values for comparison  
python log_scrubber.py fieldsummary my_fields.csv --include-raw  
  
# Disable CUI marking detection (if your data isn't CUI-regulated)  
python log_scrubber.py my_events.csv --no-cui
```

Output is written to `my_fields_scrubbed_20260420_103045.csv` (auto-timestamped).

Output Format

Fieldsummary Mode

By default, the output contains scrubbed values only - the raw values column is removed so the output is safe to share:

```
Field Name, Scrubbed Values, Count, Distinct Count
```

Use `--include-raw` to keep the original raw values alongside the scrubbed values (useful for reviewing what was changed):

```
Field Name, Raw Values, Scrubbed Values, Count, Distinct Count
```

Samples Mode

Log sample events are scrubbed in place - the output file contains only scrubbed events.

Step-by-Step Workflow

1. Discover Sourcetypes (optional)

If you don't already know which index and sourcetype to target, run this first against your Splunk metadata index:

```
| tstats count where index=* AND sourcetype={sourcetype(s)} earliest=-30d BY  
sourcetype, index, source  
| stats values(index) as indexes, values(source) as sources, sum(count) as  
event_count by sourcetype  
| sort sourcetype
```

Replace `{sourcetype(s)}` with a specific sourcetype or wildcarded filter (e.g., `cisco:asa` or `cisco:*`) and adjust the `index=...` filter for your own environment. Keep the sourcetype filter specific in large deployments or the search can run for a very long time. The `earliest=-30d` window also avoids surfacing sourcetypes that no longer exist.

Click **Export** -> choose **CSV** -> save the file (e.g., `sourcetypes.csv`).

This lists each matching sourcetype with its indexes, sources, and event counts so you can pick a target for the next two searches. **Do not scrub this export** - sourcetype and index names need to stay accurate for downstream use (CIM macros, Data Refinery *Sourcetype* CSV entry, etc.). Only redact a name by hand if it itself contains sensitive content.

2. Export Field Values from Splunk Web

Run this SPL in Splunk Web (adjust index, sourcetype, and time range):

```
index=<your_index> sourcetype="<your_sourcetype>" earliest=-7d@h latest=now  
| fieldsummary maxvals=5  
| search field!="_*" AND field!="date_*" AND field!="linecount"
```

```
AND field!="punct" AND field!="timestartpos" AND field!="timeendpos"
AND field!="splunk_server_group"
```

Click **Export** -> choose **CSV** -> save the file (e.g., `guardduty_fields.csv`).

3. Export Log Samples from Splunk Web

```
index=<your_index> sourcetype="<your_sourcetype>" earliest=-1d@d latest=now
| dedup punct | head 20
```

Click **Export** -> choose **CSV** -> save the file (e.g., `guardduty_samples.csv`).

Tip: consider dropping the `| head 20` cap (but keeping `| dedup punct`) to capture every distinct event pattern. `dedup punct` already collapses the result set to one event per unique punctuation shape, so the unbounded query usually returns only a few dozen to a few hundred rows - completely manageable for most sourcetypes. The 20-event cap can silently hide infrequent-but-important patterns (rare error variants, privileged-user actions, edge-case payloads), so for small or medium sourcetypes the unbounded form is often the better choice. Keep the cap for very high-cardinality sourcetypes where the unbounded result would be too large.

Alternatively, copy/paste raw events into a plain `.txt` file (one event per line). The scrubber auto-detects the format.

4. Scrub the Exports

Skip the discovery export from Step 1 - it should pass through untouched. Run the scrubber on the field-value and log-sample exports:

```
# Auto-detection (recommended)
python log_scrubber.py guardduty_fields.csv
python log_scrubber.py guardduty_samples.csv

# Or scrub everything from today's Downloads in one shot
python log_scrubber.py ~/Downloads/*.csv
```

5. Review and Send

Check the `*_scrubbed_*` output files to verify sensitive data was replaced, then transmit the scrubbed files to the downstream recipient (LLM pipeline, email, ticket, etc.).

Read every line before you send. The scrubber redacts what it can recognize by shape (IPs, emails, tokens, SSNs, CUI markings) or by structured field name. It **cannot** detect bare personal names, free-text content, or organization-internal terms that have no distinguishing pattern. Common blind spots: an `assigned_to` / `assignee` / `requester` / `owner` value like `John Smith` sitting in an unstructured `_raw` event, a `comment` / `description` / `notes` field containing customer names or case details, a person-named asset like `JSMITH-LAPTOP`, internal project or customer code names, branch and team names, and

stack traces that echo user input. If a sensitive value lives in its own JSON key, CSV column, or fieldsummary field, you can target it with an `@json,<field_name>,<replacement>` rule and re-run - the `@json` shortcut covers all three. If it's buried in narrative text, no automated tool can find it reliably. Your eyes are the control of last resort.

Configuration

Config File Location

The scrubber automatically looks for `log_scrubbing_config.csv` in these locations (first match wins):

1. Current working directory: `./log_scrubbing_config.csv`
2. Data subdirectory: `./data/log_scrubbing_config.csv`
3. Same directory as the script: `<script_dir>/log_scrubbing_config.csv`
4. Data subdirectory of script: `<script_dir>/data/log_scrubbing_config.csv`

The simplest approach is to place the config file in the same folder as `log_scrubber.py`. You can also specify a config explicitly:

```
python log_scrubber.py fieldsummary fields.csv --config /path/to/my_rules.csv
```

Built-in Scrubbing (Always Active)

Even without a config file, the scrubber applies these regex patterns automatically:

Pattern	Replacement
IPv4 addresses (<code>192.168.1.50</code>)	<code>10.0.0.x</code>
AWS ip- hostnames (<code>ip-10-50-26-117</code>)	<code>ip-10-0-0-x</code>
Email addresses (<code>admin@corp.com</code>)	<code>user@example.com</code>
FQDN hostnames (<code>server01.company.com</code>)	<code>host.example.com</code>
UNC paths (<code>\\server\share</code>)	<code>\\SERVER\SHARE</code>
Domain usernames (<code>CORP\jsmith</code>)	<code>DOMAIN\user</code>
MAC addresses (<code>00:1A:2B:3C:4D:5E</code>)	<code>00:00:00:00:00:00</code>
US Social Security Numbers (valid ranges only)	<code>000-00-0000</code>
Luhn-validated credit cards	<code>4111-1111-1111-1111</code>
US phone numbers	<code>555-555-5555</code>
Windows user SIDs (<code>S-1-5-21-*</code>)	<code>S-1-5-21-0-0-0-0</code>
AWS access keys (<code>AKIA...</code> , <code>ASIA...</code>)	<code>AKIAREDACTEDREDACTED</code>
JSON Web Tokens (<code>eyJ...</code>)	<code>eyJREDACTED.REDACTED.REDACTED</code>

Pattern	Replacement
GitHub PATs (<code>ghp_*</code> , <code>gho_*</code> , etc.)	<code>ghp_REDACTED</code>
Slack tokens (<code>xoxb-*</code> , <code>xoxp-*</code> , etc.)	<code>xoxb-REDACTED</code>
Stripe keys (<code>sk_live_*</code> , <code>sk_test_*</code>)	<code>sk_test_REDACTED</code>
Google API keys (<code>AIza*</code>)	<code>AIzaREDACTED</code>
PEM private key blocks	<code>[REDACTED_PRIVATE_KEY]</code>
HTTP Authorization headers	<code>Authorization: <scheme> REDACTED</code>
URL query-string credentials	<code>password=REDACTED</code> , etc.
US National Provider Identifiers (Luhn-validated)	<code>1234567893</code>

CUI markings (CMMC / NIST SP 800-171) trigger whole-line redaction with a placeholder that preserves the marking category and original byte count. See the README for the full list of detected marking styles.

Config File Format

The config file is a CSV with four types of rules:

Text substitution - replaces literal strings anywhere in the data:

```
# Simple replacement
sensitive-hostname,single,REDACTED_HOST

# Random replacement (picks one each time)
my-secret-domain.com,random,"example1.com,example2.com,example3.com"

# Two-column shorthand (implicit single mode)
my-company.com,example.com
```

JSON field rules (`@json` prefix) - targets specific field names at any nesting depth in JSON data:

```
@json,accessKeyId,REDACTED_KEY
@json,accountId,random,"000000000001,000000000002,000000000003"
@json,userName,REDACTED_USER

# Qualified path - matches "arn" ONLY when nested under userIdentity
@json,userIdentity.arn,arn:aws:iam::REDACTED:user/REDACTED
```

Key-value tag matching - handles AWS/Azure/GCP tag structures like `{"key": "Owner", "value": "admin@corp.com"}`:

```
# Use the tag's key name (not "key" itself)
@json,Owner,random,"user_a@example.com,user_b@example.com"
```

```
@json,Environment,REDACTED_ENV
```

Bulk name lists (@names prefix, v1.3.1+) - one config row expands at load time into one text rule per listed name, so an employee/contractor roster doesn't need a separate row per person. Supports both **single** and **random** replacement modes, with the same semantics as ordinary text rules (in **random** mode each occurrence picks independently, so a given source name may map to different replacements at different points in the file):

```
# Every listed name maps to the same replacement
@names,"Alice Smith,Bob Jones,Carol Davis",single,REDACTED_NAME

# Each occurrence picks randomly from the replacement pool
@names,"Thomas Kincade,Mary Bruce,Joe Wadamaker",random,"John Doe,Jane Doe,Joe Smith"
```

The names list is just a quoted CSV field, so it can be broken across lines for readability:

```
@names,"Thomas Kincade,
Mary Bruce,
Joe Wadamaker",random,"John Doe,Jane Doe,Joe Smith"
```

Lines starting with # are treated as comments.

Example Config File

```
# === Text Substitution Rules ===
acme-corp.com,single,example.com
acmecorp,single,examplecorp
my-splunk-server,single,splunk-host

# === JSON Field Rules ===
@json,accessKeyId,REDACTED_KEY
@json,accountId,random,"000000000001,000000000002,000000000003"
@json,userName,REDACTED_USER
@json,sourceIPAddress,10.0.0.x
@json,Owner,random,"user_a@example.com,user_b@example.com"
@json,Name,REDACTED_NAME
@json,Environment,random,"dev,staging,prod"

# === Bulk Name Lists ===
@names,"Alice Smith,Bob Jones,Carol Davis",random,"John Doe,Jane Doe,Joe Smith"
```

Command Reference

```
usage: log_scrubber.py [-h] [--config CONFIG] [--output OUTPUT]
                      [--no-builtin] [--no-cui] [--include-raw]
                      [--dry-run] [--quiet] [--version]
                      [mode] input [input ...]
```

Argument	Description
<code>mode</code>	Optional. Either <code>fieldsummary</code> or <code>samples</code> . Omit for auto-detection.
<code>input</code>	One or more input file paths. Glob patterns (<code>*.csv</code>) are expanded.
<code>--config, -c</code>	Path to scrubbing config CSV (auto-detected if not specified)
<code>--output, -o</code>	Output file path (single input only; ignored in batch mode). Default: <code><input>_scrubbed_<timestamp>.<ext></code>
<code>--no-builtin</code>	Disable built-in regex patterns (IPs, emails, tokens, SSN, CC, etc.)
<code>--no-cui</code>	Disable CUI marking detection (enabled by default)
<code>--include-raw</code>	Include the original raw values column in fieldsummary output (default: scrubbed only)
<code>--dry-run</code>	Show what would be done without writing output
<code>--quiet, -q</code>	Suppress informational output
<code>--version</code>	Print version and exit

Supported Input Formats

Fieldsummary Mode

Accepts two CSV formats (auto-detected):

1. **Splunk fieldsummary export** - columns: `field`, `count`, `distinct_count`, `values`, etc.
2. **Pre-scrubbed export** - columns: `Field Name`, `Raw Values`, `Scrubbed Values`, `Count`, `Distinct Count`. This is the same format the scrubber itself writes with `--include-raw`, so re-running the tool on its own output is supported.

Both formats are handled automatically. Output column headers are normalized to `Field Name` and `Scrubbed Values` regardless of input format.

Samples Mode

Handles three formats (auto-detected):

1. **CSV with `_raw` column** - standard Splunk CSV export. The `_raw` column is scrubbed in place.
2. **Plain text** - one event per line (syslog, key=value, etc.)
3. **JSON / JSONL** - single-line or multi-line JSON events. Brace depth tracking handles multi-line JSON objects that span multiple lines.

Library Use

`log_scrubber.py` is import-safe - the CLI banner and argument parsing are guarded by `if __name__ == "__main__"`, so `import log_scrubber` from another module has no side effects.

Class API (recommended for new code):

```
from log_scrubber import Scrubber

scrubber = Scrubber(config_path='log_scrubbing_config.csv')
safe_text = scrubber.scrub(raw_log_line)

# Batch convenience
safe_batch = scrubber.scrub_many([line1, line2, line3])

# Programmatic construction with inline rules
scrubber = Scrubber(
    text_rules=[('acme.com', 'single', 'example.com')],
    json_field_rules=[('accountId', 'single', '000000000000'),
                      ('userIdentity.arn', 'single',
                       'arn:aws:iam::REDACTED:user/REDACTED')],
    enable_builtins=True,
    enable_cui=True,
)
```

A single `Scrubber` instance is safe to share across threads as long as the rule lists aren't mutated after construction.

Functional API (backward compatible):

```
from log_scrubber import parse_scrubbing_config, scrub_text

text_rules, json_rules = parse_scrubbing_config('log_scrubbing_config.csv')
clean = scrub_text(raw, text_rules, json_rules)

# Optional flags (default True) for v1.1+ feature control
clean = scrub_text(raw, text_rules, json_rules,
                   enable_builtins=True, enable_cui=True)
```

Existing callers that don't pass the new kwargs get the v1 behavior plus the new scrubbing layers automatically.

Tips

Reuse configs across tools. The same `log_scrubbing_config.csv` works with both `log_scrubber.py` and the Paydirt browser tool. If you maintain a config file for one, the other will use it as-is.

Review output before sending. Automated scrubbing handles known patterns, but always spot-check the output for any environment-specific data the rules might have missed.

Large JSON events (GuardDuty, CloudTrail, etc.) are handled natively - the scrubber parses JSON at any nesting depth and applies `@json` rules recursively, including AWS/Azure/GCP tag structures.

Add rules incrementally. Start with the built-in patterns, review the output, then add `@json` and text rules to the config for environment-specific identifiers that slipped through.

Try the demo. The `paydirt_demo.log` file in the project root exercises every scrubbing category. Run it through the scrubber to see all the built-in patterns in action:

```
python log_scrubber.py paydirt_demo.log
```

Version History

For the full version history, see the [Version History section in the project README](#).

License

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